

Embeddings and retrieval augmented generation

Foundations of AI

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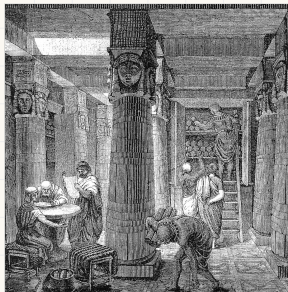
26 September 2026

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Section 1

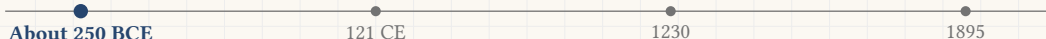
Indexes before computers

Alexandria, about 250 BCE



Imagined by O. Von Corven, 19th century.

- A scroll has no title page. To learn what it is, you unroll it.
- **A scholar arrives wanting the works of one author.**
- Callimachus wrote **one entry per work: author, opening words, length in lines**, in 120 scrolls.
- You read the entries until you know which of **the scrolls** to unroll.



China, 121 CE



- A writing system with no alphabet has no order to sort by.
- **You meet a character you cannot read.**
- Xu Shen gave each one an address: **one of 540 components**, and after 1615 **a stroke count too**.
- Follow the address and you reach **the entry: what it means, how it is said.**

A page of the Kangxi Dictionary, 1716.

About 250 BCE

121 CE

1230

1895

Paris, 1230



Hugh of Saint-Cher. Treviso, 1352.

- About 500 friars listed every word of the Bible, and everywhere it appears.
- **You want every passage that uses one word.**
- There was no way to say where, so they cut every chapter into **seven parts, marked A to G.**
- The address sends you to **the passages themselves.** We still cite that way.

About 250 BCE

121 CE

1230

1895

Brussels, 1895



The Repertoire, about 1900.

- Otlet and La Fontaine began **one index card for every book and article ever printed**: 400,000 in the first year, later over 15 million.
- **From 1896 you could post them a question, for a fee.**
- They searched the cards and mailed back the ones that matched.
- The cards told you where to find **the books and the articles**. Over 1,500 questions a year by 1912.

About 250 BCE

121 CE

1230

1895

The same three things, four times

	Query	Keys	Values
Alexandria, 250 BCE	The works of one author	Author, opening words, length in lines	The scrolls
China, 121 CE	A character you cannot read	A component and a stroke count	The dictionary entry
Paris, 1230	Every passage using a word	Book, chapter, letter A to G	The passages
Brussels, 1895	A question sent by post	Fifteen million index cards	The books and articles
A company, 2026	A question in plain words	?	The company's own documents

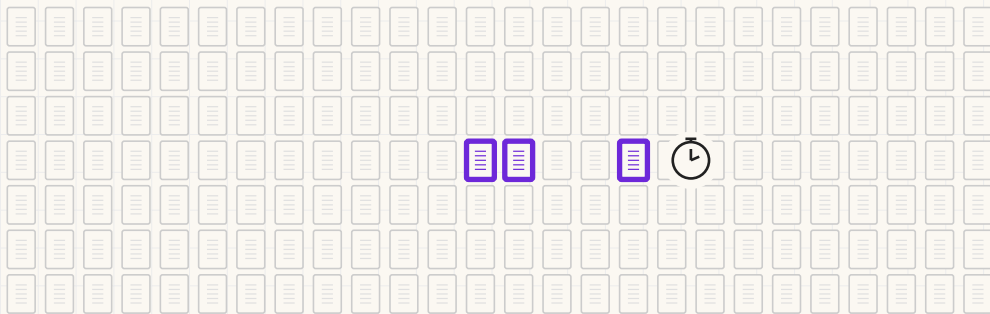
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Section 2

Information retrieval

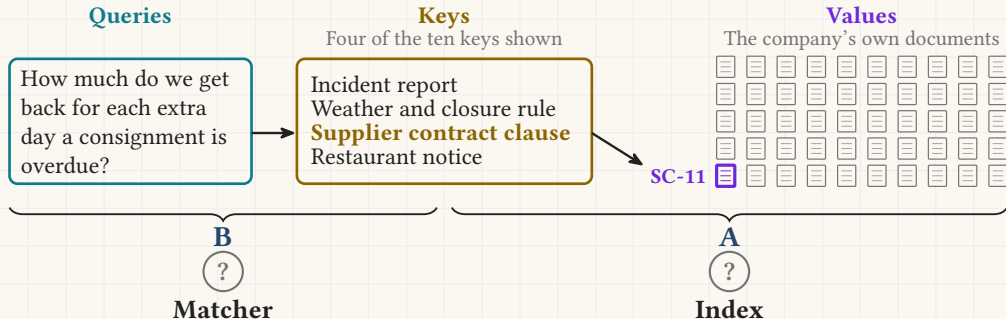
Finding the few documents that answer a question

One company's own documents



“What did we agree with the rope supplier about late delivery?”

Queries, keys and values



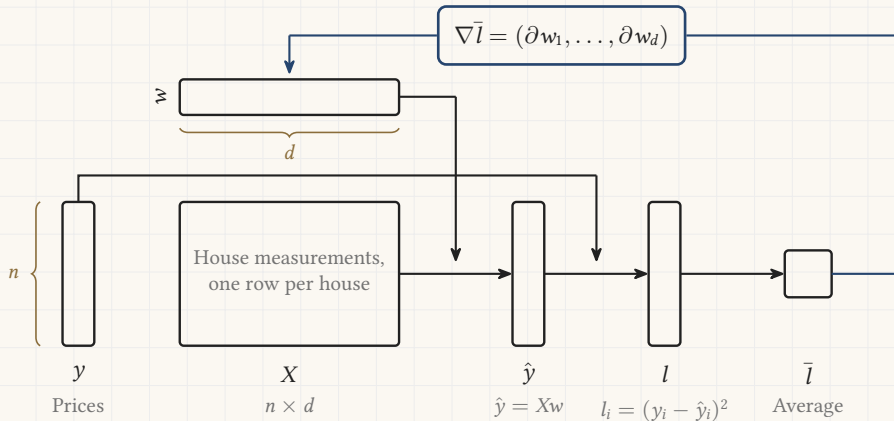
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Section 3

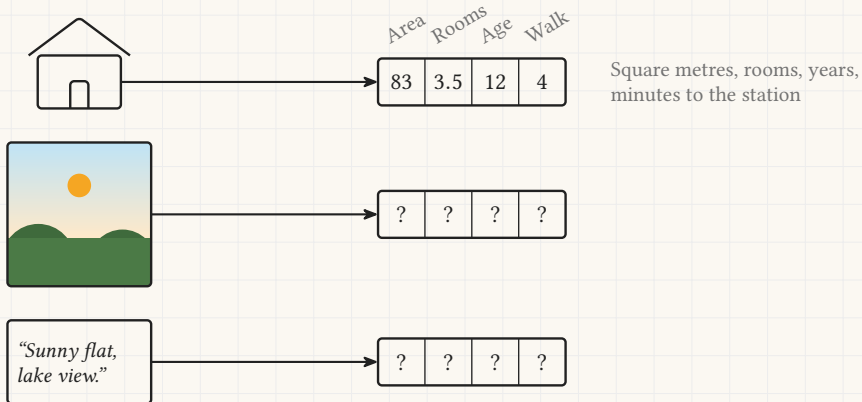
Machine learning needs vectors

Linear regression with gradient descent

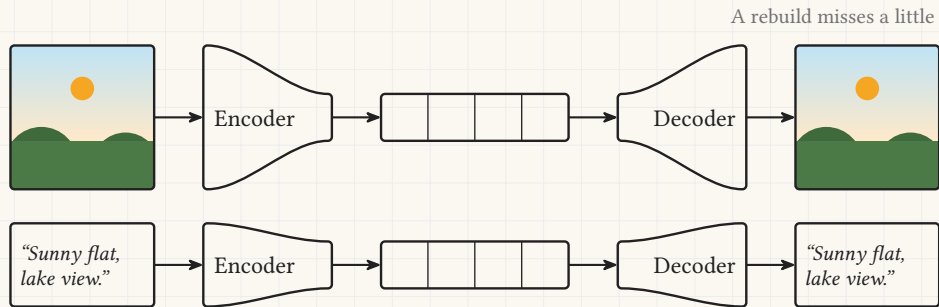
At one o'clock, in one picture. Each box is a piece of data, and its shape is the shape of the data.



How do we do machine learning with images and text?



Autoencoders



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Section 4

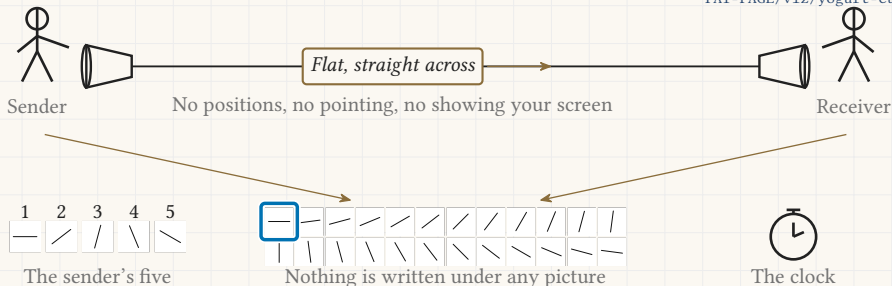
The yogurt cup game

The yogurt cup game

- Both see every picture
- One sees five of them, in order
- Only words go through the cup
- The partner finds them, against the clock



<https://carloscotrini.github.io/FAI-PAGE/viz/yogurt-cups/>



The score is the time

Round 1

Steep, nearly upright



"Not that one. The clock is still running."



The clock

Between rounds

*"The pairs who get fast are the ones
who stop describing and start counting."*

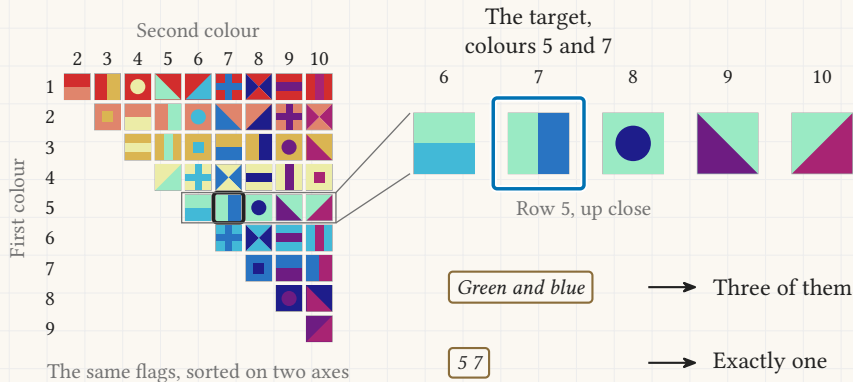
Round 2

Ten steps up from flat

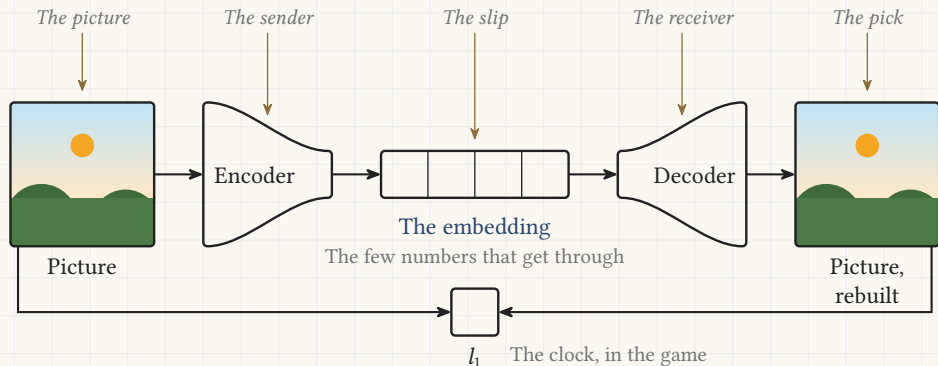


Right, first time

Words are a coarse code



The game, named

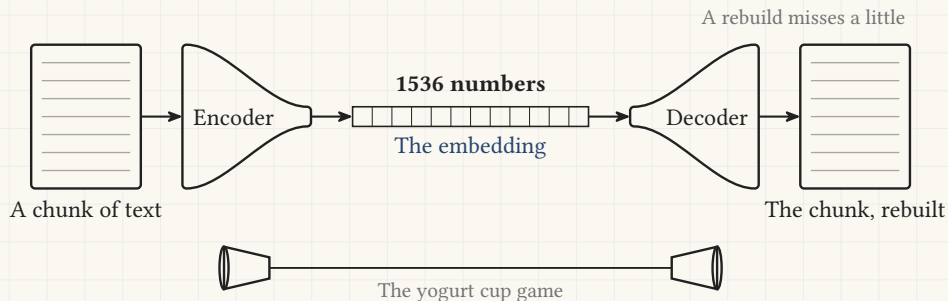


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Section 5

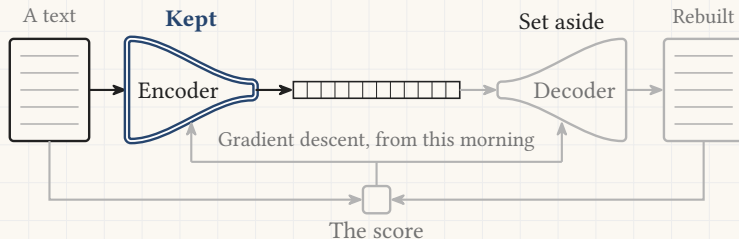
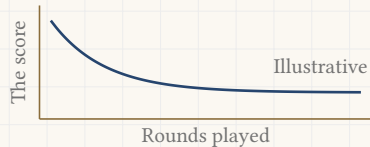
Embeddings of text

An autoencoder for text

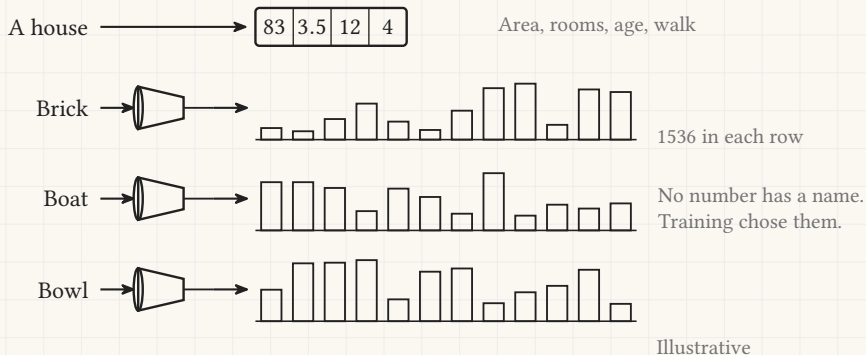


How it is trained

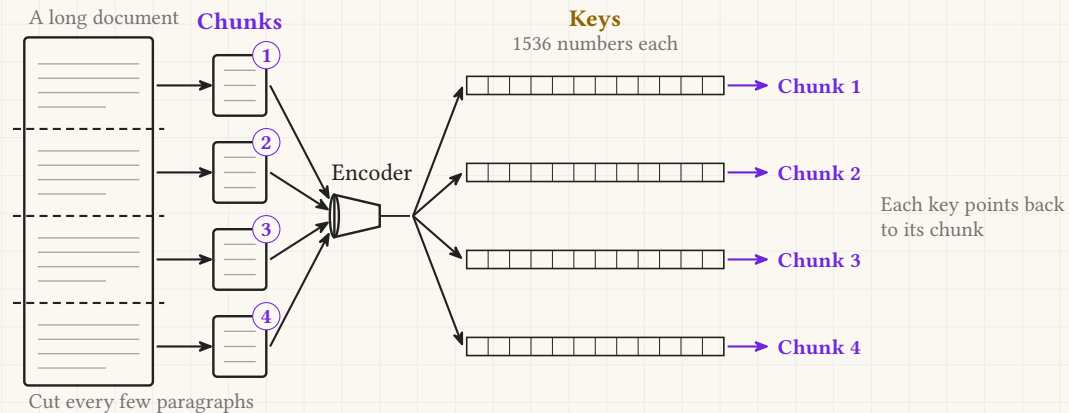
- 1 Collect a large collection of texts
- 2 Initialise the weights at random
- 3 Play the game, take the score
- 4 Update the weights from the score, play again



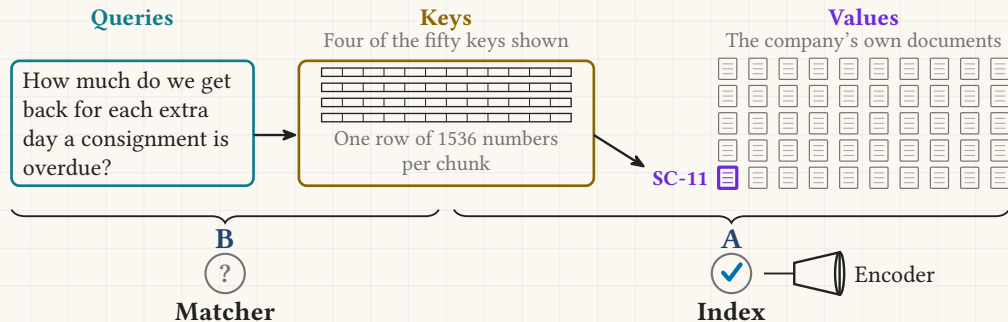
The numbers have no names



One key per chunk



Queries, keys and values

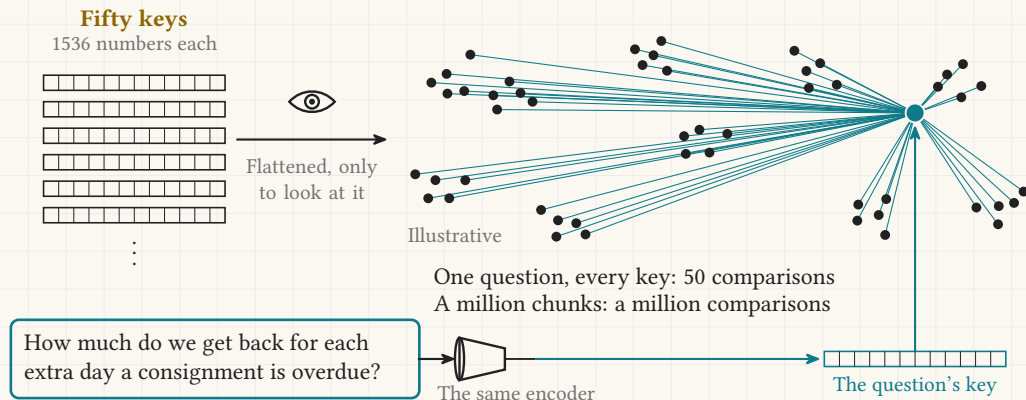


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Section 6

Finding the relevant chunks

How do we effectively find the relevant keys?



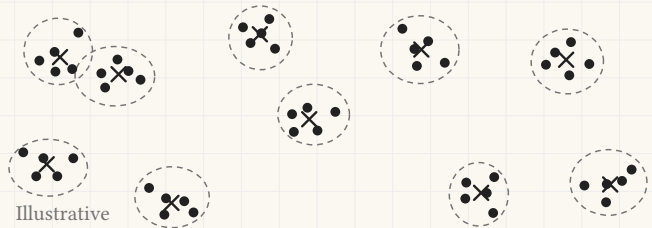
Done once, before any question

Yesterday's k-means

One centre per clump



Each centre keeps the list of its chunks



At question time

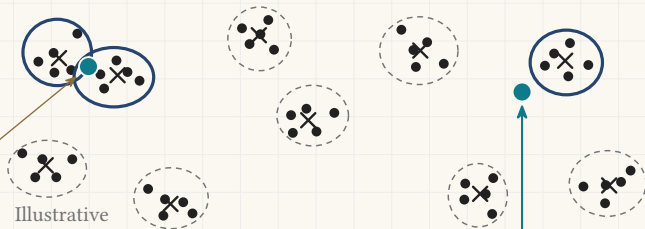
The chunks behind the closest keys



SC-11

15 comparisons instead of 50

A question near a border: its answer can sit in the next clump.
Real systems open more than one.



Illustrative

10 comparisons, one per centre
5 comparisons, inside that clump

How much do we get back for each extra day a consignment is overdue?



The same encoder



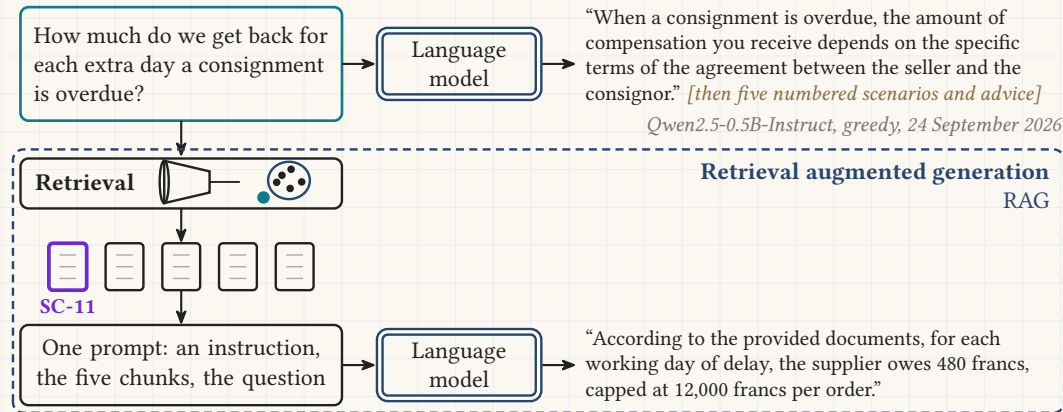
The question's key

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Section 7

Retrieval augmented generation

The full system



The prompt we send to the model

Written once, by a person

Answer the question using only the documents below. If the documents do not contain the answer, say so.

Found by retrieval

Document 1. Kernli Seiltechnik AG, framework agreement 2024.
Clause 6.2. Replacement haul ropes are delivered to Vorlen Dorf within 21 working days of a written order. **For each working day of delay the supplier owes 480 francs, capped at 12,000 francs per order.** Clause 6.3. The warranty on splicing work is 36 months from handover. Clause 6.4. Orders above 50,000 francs need a second signature.

[documents 2 to 5, four more chunks]

Asked by the user

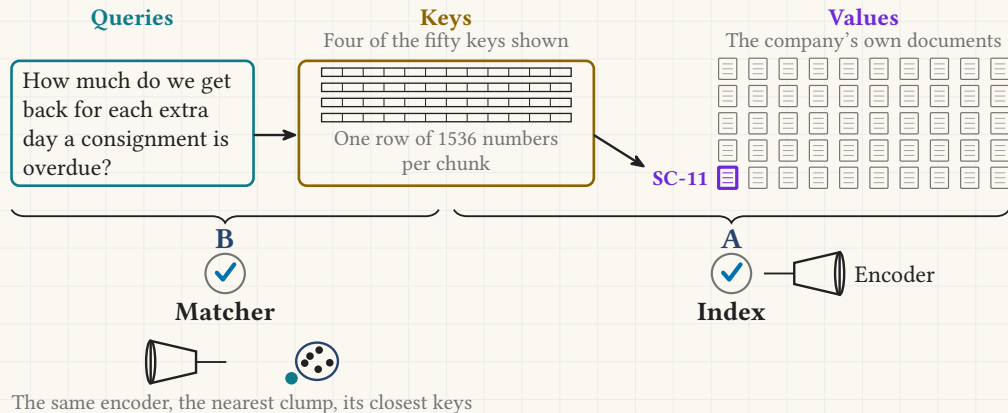
Question: How much do we get back for each extra day a consignment is overdue?

What the model wrote

According to the provided documents, for each working day of delay, the supplier owes **480 francs, capped at 12,000 francs per order.**

*Qwen2.5-0.5B-Instruct, greedy,
24 September 2026*

Queries, keys and values



References: indexes before computers

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